Type MS · Describe fertilisation and early embryo development, including twins

Answer 1 of 7 · Question 6 part b · 4 marks

2023 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 2 of 7 · Question 5 part a.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

5	(a)	(i)	ovulation;		1
		(ii)	fertilisation		1
		(iii)	Any two from: • zygote/embryo/cells divides/splits; • each half develops into one offspring;		1 1
	(b)	(i)	**A** FSH and LH;		1
		(ii)	Any four from: • removal of egg from female ovary; • collect sperm from male; • egg fertilised; • embryos formed; • embryo inserted into uterus/female;	Allow egg and sperm fuse Allow zygote formed	Max 4

Total question 5 = 9 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 2 of 7 · Question 5 part a.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 3 of 7 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

5	(a)	(i)	ovulation;		1
		(ii)	fertilisation		1
		(iii)	Any two from: • zygote/embryo/cells divides/splits; • each half develops into one offspring;		1 1
	(b)	(i)	**A** FSH and LH;		1
		(ii)	Any four from: • removal of egg from female ovary; • collect sperm from male; • egg fertilised; • embryos formed; • embryo inserted into uterus/female;	Allow egg and sperm fuse Allow zygote formed	Max 4

Total question 5 = 9 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 3 of 7 · Question 5 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 4 of 7 · Question 5 part a.iii · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

5	(a)	(i)	ovulation;		1
		(ii)	fertilisation		1
		(iii)	Any two from: • zygote/embryo/cells divides/splits; • each half develops into one offspring;		1 1
	(b)	(i)	**A** FSH and LH;		1
		(ii)	Any four from: • removal of egg from female ovary; • collect sperm from male; • egg fertilised; • embryos formed; • embryo inserted into uterus/female;	Allow egg and sperm fuse Allow zygote formed	Max 4

Total question 5 = 9 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 4 of 7 · Question 5 part a.iii · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 5 of 7 · Question 8 part b.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
8	(a)	(i)	triggers ovulation/causes release of the egg;		1
		(ii)	Three from: • FSH stimulates growth/production of follicles in the ovary; • more eggs mature/produced; • increases the chances of more than one egg being released; • each egg is fertilised by a sperm;		3
	(b)	(i)	Three from: • placenta supplies glucose/oxygen from the mother; • twins have to share the supply of glucose/receive less glucose/oxygen; • less aerobic respiration; • less energy is released in the foetus and less growth occurs;	Allow nutrients for mp's 1 and 2	3
		(ii)	Two from • same egg and sperm used; • zygote splits/divides • forming 2 embryos;		2
	(c)	(i)	75th		1
		(ii)	• baby is weighed regularly; • growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age; • detect if the baby gains or loses too much mass;		3

Total 13 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 5 of 7 · Question 8 part b.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Describe fertilisation and early embryo development, including twins

Answer 6 of 7 · Question 6 part a · 2 marks

2023 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

6 (a)	• fertilisation; • egg and sperm fuse/gametes/join;		2
(b)	Any four from • mitosis; • DNA is replicated; • chromosomes move to the centre of the cell; • chromosomes are separated to opposite poles; • cell divides;	Allow one mark for prophase, metaphase, anaphase, telophase in the correct order Allow zygote/embyro divides	4
(c)	**D;** (the cells are able to differentiate into any type of body cell)		1
(d)	• (corpus luteum) produces progesterone; • to maintain uterus lining;		2
(e)	Any two from: • umbilical cord established; • transports nutrients/named nutrients/oxygen (to fetus/embryo); • removes waste products/named waste product (from fetus/embryo);		2
(f)	development of male sexual characteristics/genitalia;		1
(g)	**C;** (release of oxytocin from the pituitary gland)		1

Total for Question 6 = 13 marks

Type MS · Describe fertilisation and early embryo development, including twins

Answer 7 of 7 · Question 2 part a.iii · 1 mark

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a) (i)

Part	Role		
	where sperm are deposited	where eggs are produced	where fertilisation takes place
fallopian tube			✓
ovary		✓	
vagina	✓		

one mark for each correct row

(3)

(ii)

- muscles/uterus wall contract;
- to widen/dilate cervix;

allow description of widening cervix

(2)

(iii)

C (a sperm cell fuses with an egg cell to form a zygote)

The following statements are incorrect because:

A haploid cells are not produced from fertilisation
B an embryo is not formed from fertilisation
D a baby is not formed at fertilisation

(1)

(iv)

luteinising hormone/LH;

(1)

Type MS · Describe fertilisation and early embryo development, including twins

Answer 7 of 7 · Question 2 part a.iii · 1 mark

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

(b) (i)		One mark for each diagram	(2)
(ii)	mitosis → two cells, each with 23 chromosomes mitosis → four cells, each with 46 chromosomes meiosis → two cells, each with 46 chromosomes meiosis → four cells, each with 23 chromosomes	reject more than one line from each type of cell division	(2)

Total for Question 2 = 11 marks